

Regularization / Normalization / Init. for NN

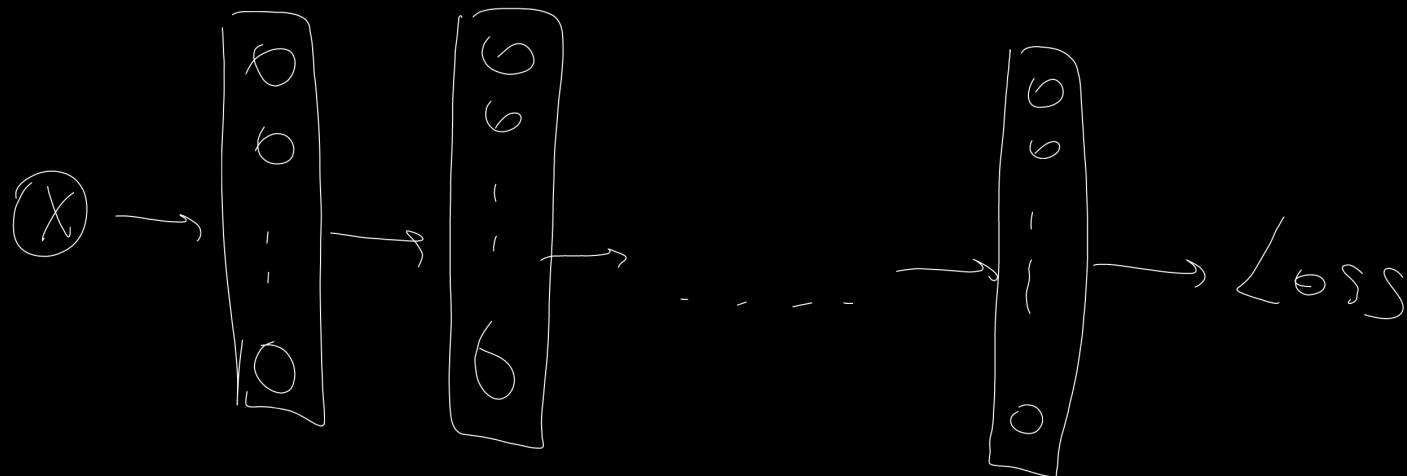
L_2 -norm. (weight decay)

$$\frac{1}{N} \sum_{i=1}^N \mathcal{L}(y_i, f(x_i, \theta)) + \frac{\lambda}{2} \|\theta\|_2^2 \rightarrow \min_{\theta}$$

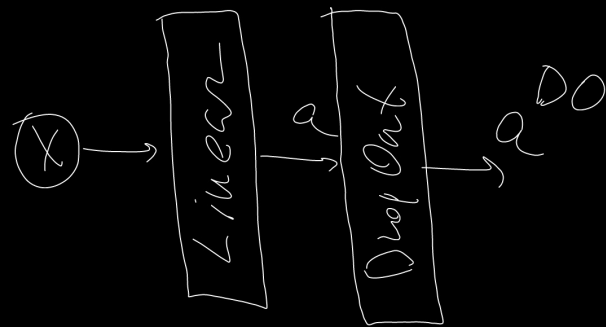
$$\left\{ \begin{array}{l} \frac{1}{N} \sum_i \mathcal{L}(y_i, \hat{f}(x_i, \theta)) \rightarrow \min_{\theta} \\ \|\theta\|_2^2 \leq \eta \end{array} \right.$$

Apply L_2 for weights, but not biases

MLP



Drop Out



model.train()
model.eval()

$$z = Wx + b$$

$$a = g(z)$$

$$y_i \sim \text{Bern} : \begin{matrix} 0 & 1 \\ p & 1-p \end{matrix}$$

$$a_i^{DO} = a_i \cdot y_i$$

$$E_{y_i \sim \text{Bern}} a_i^{DO} = p(0 \cdot a_i) + (1-p)(1 \cdot a_i) = (1-p)a_i$$

Classic DO

$$\text{Train: } y_i \sim \text{Bern}$$

$$a_i^{DO} = y_i \cdot a_i$$

$$\text{Test: } a_i^{DO} = (1-p) \cdot a_i$$

Inverted DO

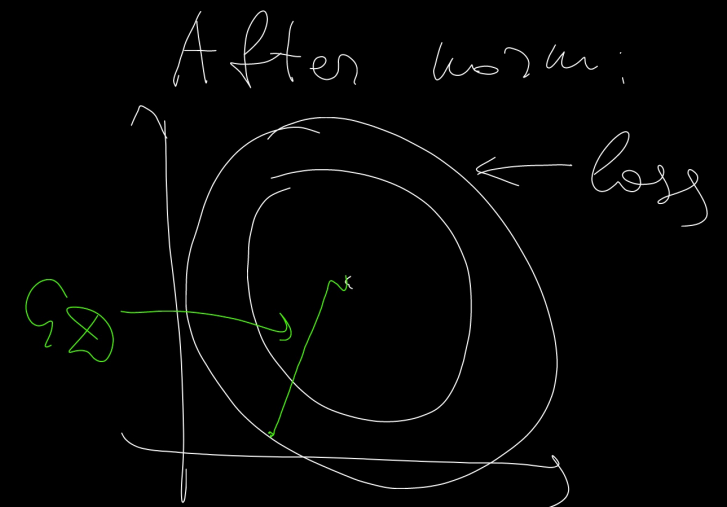
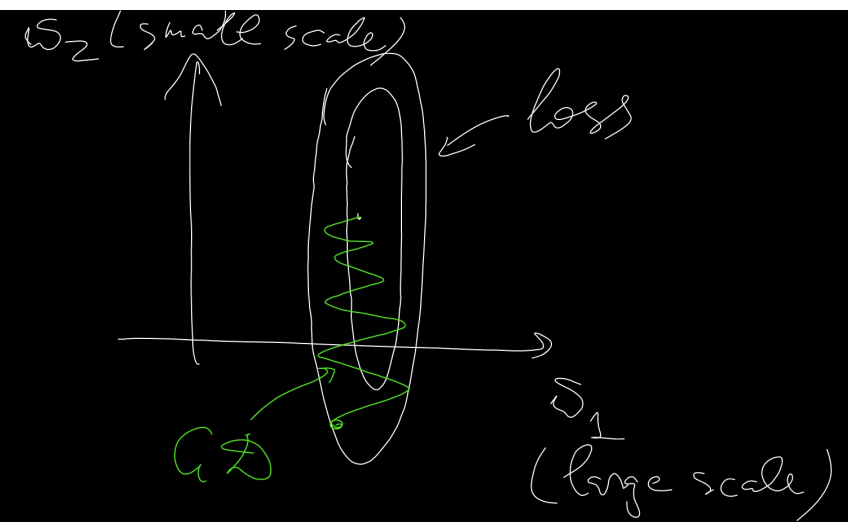
$$\text{Train: } y_i \sim \text{Bern}$$

$$a_i^{DO} = \frac{y_i \cdot a_i}{1-p}$$

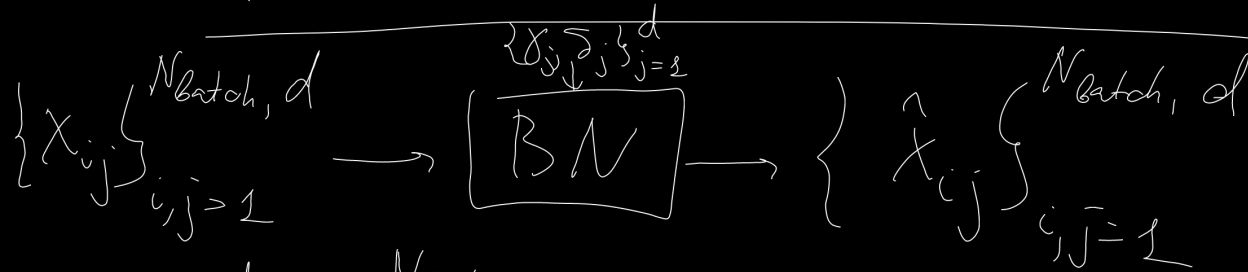
$$\text{Test: } a_i^{DO} = a_i$$

Normalization

- faster optimization
- better regularization
- better comp. stability



Batch Normalization



Train: comp. μ_j, σ_j for current mini-batch and learn $\{\mu_j, \sigma_j\}$

Test: use EMA for μ_j, σ_j

$$\mu_j = \frac{1}{N_{batch}} \sum_{i=1}^{N_{batch}} x_{ij} \quad \forall j=1 \dots d$$

$$\sigma_j^2 = \frac{1}{N_{batch}} \sum_i (x_{ij} - \mu_j)^2$$

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}$$

$$\hat{x}_{ij} = \gamma_j z_{ij} + \delta_j$$



BN can't be used for small N_{batch}

Alternative Normalizations for Images

image $x \in \mathbb{R}^{H \times W \times C}$
batch $\in \mathbb{R}^{B \times H \times W \times C}$

Batch Norm: for each C average w.r.t. B, H, W

Layer Norm: for each B average w.r.t. H, W, C

Instance Norm: for each B, C average w.r.t. H, W

Normalization for texts

Text $x \in \mathbb{R}^{T \times d}$
Batch $\in \mathbb{R}^{B \times T \times d}$

not used for
texts!

Batch Norm: for each d average w.r.t. B, T

Layer Norm: for each B average w.r.t. T, d
or for each T average w.r.t. d

RMS norm: for each B, T average w.r.t. d

$$x = \{x_i\}_{i=1}^d \quad \text{RMS}(x) = \sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2 + \epsilon}, \quad \hat{x}_i = \frac{x_i}{\text{RMS}(x)} \cdot \gamma_i$$

Weights initialization

$$z = Wx, \quad b_{\text{start}} = 0$$
$$\nabla_x f = W^T \nabla_z f$$

$$\frac{\partial f}{\partial x_j} = \sum_{i=1}^{n_{\text{out}}} (W_{ji}^T) \frac{\partial f}{\partial z_i}$$

$$\text{Var}\left(\frac{\partial f}{\partial z_i}\right) = 1$$

$$\Rightarrow E \frac{\partial f}{\partial x_j} = 0, \quad \text{Var}\left(\frac{\partial f}{\partial x_j}\right) = n_{\text{out}} \cdot \text{Var}(W)$$

$$W_{ij} \sim \mathcal{N}(0, \text{Var}(W)), \quad x_j \sim \mathcal{N}(0, 1)$$

$$z_i = \sum_{j=1}^{n_{\text{in}}} W_{ij} x_j; \quad E z_i = E \sum_{j=1}^{n_{\text{in}}} W_{ij} x_j = \sum_{j=1}^{n_{\text{in}}} E W_{ij} E x_j = 0$$

$$\text{Var}(z_i) = \sum_{j=1}^{n_{\text{in}}} \text{Var}(W_{ij}) \text{Var}(x_j) = n_{\text{in}} \text{Var}(W) = 1 \Rightarrow \text{Var}(W) = \frac{1}{n_{\text{in}}}$$

Xavier (Glorot) initialization

$$\text{Var}(W) = \frac{2}{n_{in} + n_{out}}$$

Apply for sigmoid/tanh activations

$$W_{ij} \sim \mathcal{N}(0, \text{Var}(W))$$

$$W_{ij} \sim \mathcal{R}\left[-\sqrt{\frac{6}{n_{in} + n_{out}}}, \sqrt{\frac{6}{n_{in} + n_{out}}}\right]$$

Kaiming (He) initialization

$$y = \max(0, z) \quad , \quad E y^2 = \int_{-\infty}^{\infty} \max(0, z)^2 p(z) dz =$$
$$\text{Var}(w) = \frac{2}{n_{in}} = \int_0^{\infty} z^2 p(z) dz = \frac{1}{2} E z^2$$

$$w_{ij} \sim \mathcal{N}(0, \text{Var}(w))$$

$$w_{ij} \sim \mathcal{R}\left[-\sqrt{\frac{6}{n_{in}}}, \sqrt{\frac{6}{n_{in}}}\right]$$

Apply for ReLU,
SiLU, ELU

SGD

$$F(w) = \frac{1}{N} \sum_{i=1}^N f_i(w) \rightarrow \min_w, \quad N \gg 1$$

$$\nabla F(w) = \frac{1}{N} \sum_{i=1}^N \nabla f_i(w) \quad \text{-- too costly}$$

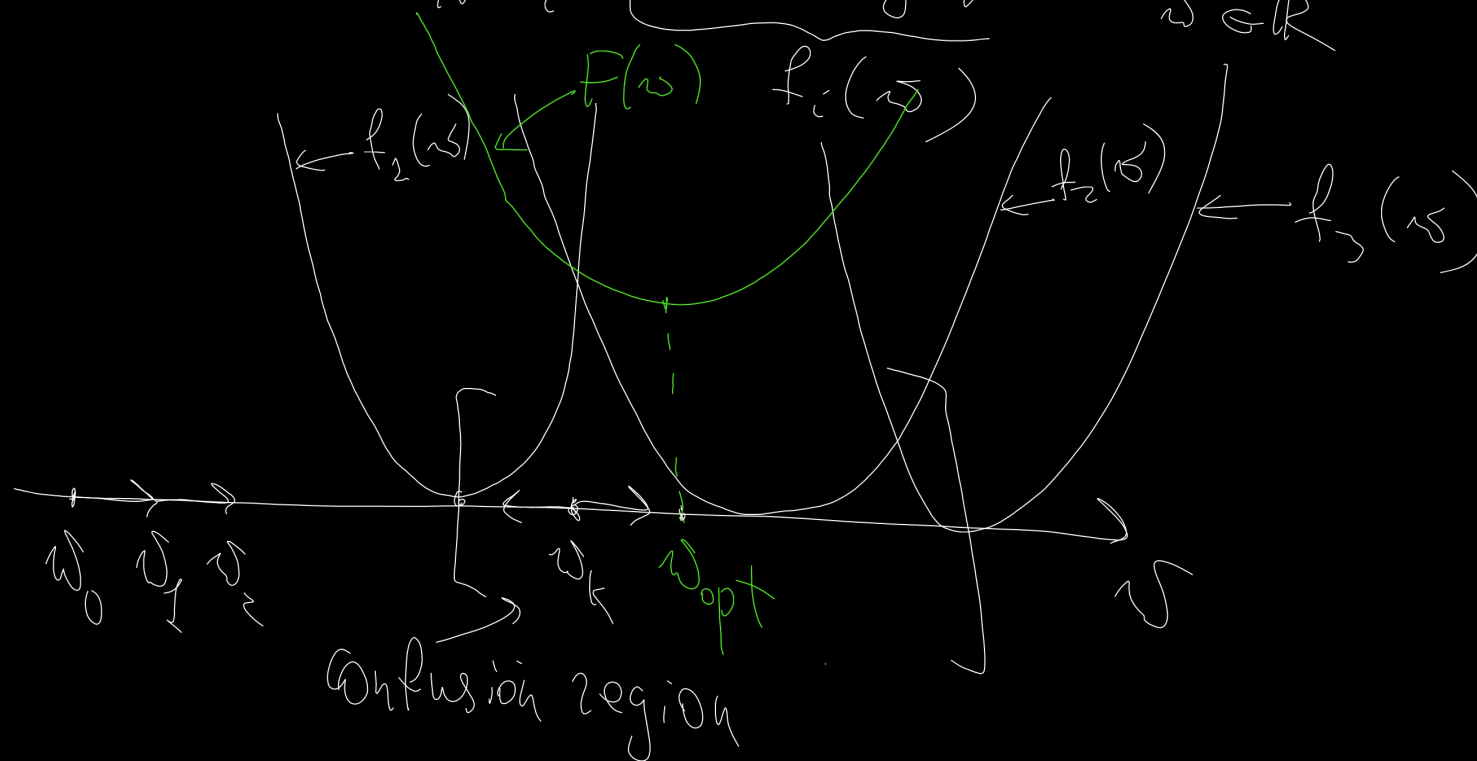
$$\text{SGD: } \begin{cases} I_k \subset \text{Unif}(1, 2, \dots, N) \\ g_k = \frac{1}{|I_k|} \sum_{i \in I_k} \nabla f_i(w_k) \\ w_{k+1} = w_k - \alpha_k g_k \end{cases}$$

Example: 1d linear reg.

$$F(w) = \frac{1}{N} \sum_i (wx_i - y_i)^2 \rightarrow \min_{w \in \mathbb{R}}$$

In SGD
for convergence

$$d_k \rightarrow 0$$



SGD as noisy full GD

$$\hat{g}_k := \frac{1}{N} \sum_{i=1}^N \nabla f_i(w_k)$$

$$g_k = \frac{1}{|I_k|} \sum_{i \in I_k} \nabla f_i(w_k) \sim \mathcal{N}(g_k, \Sigma_k)$$

$$w_{k+1} = w_k - \alpha_k g_k = w_k - \alpha_k \left(\hat{g}_k + \varepsilon_k \right), \quad \Sigma_k \rightarrow 0 \text{ if } |I_k| \rightarrow N, \quad \varepsilon_k \sim \mathcal{N}(0, \Sigma_k)$$

